



NATIONAL UNIVERSITY OF MODERN LANGUAGES

MOBILE APPLICATION DEVELOPMENT LAB REPORT – 5

NAME: Abdul Rafay

ROLL NO: FL23791

PROGRAM: BSSE (6th Semester)

COURSE: Mobile Application Development

SUBMITTED TO: Sir M Arif

DATE: 22 February 2026

DAY: Sunday

Variable Declaration Types

1. Using Variables

```
void main() {  
    var name = "Abdul Rafay";  
    var age = 22;  
    print(name);  
    print(age);  
}
```

Output

```
Abdul Rafay  
22
```

2. Explicit Type Declaration

```
void main() {  
    String name = "Rafayeee";  
    int age = 13;  
    bool isAdult = true;  
    double area = 13.34;  
    print(name);  
    print(age);  
    print(area);  
    print(isAdult);  
}
```

```
Rafayeee  
13  
13.34  
true
```

3. Using Dynamic Type

```
void main() {  
    dynamic value = "Beautiful";  
    value = 12;  
    print(value);  
}
```

```
12
```

Built-in Datatypes

1. Numbers

```
void main() {  
    // Integer datatype  
    int age = 23;  
    // Double datatype  
    double area = 10 / 3;  
    print(age);  
}
```

```
print(area);  
}
```

23

3.3333333333333335

2. String

```
void main() {  
    // Single line string  
    String name = "rafay";  
    // Multi line string  
    String data = "Hi hello world gee";  
    // String concatenation  
    int age = 12;  
    print(name);  
    print(data);  
    print("My age: ${age + 3}");  
}
```

```
rafay  
Hi hello world gee  
My age: 15
```

3. Lists (Arrays)

```
void main() {  
    var fruits = ["Apple", "Orange", "Banana"];  
    var number = [1, 2, 3, 4, 5];  
    fruits.add("Grapes");  
    fruits.insert(1, "Strawberry");  
    print(fruits);  
    fruits.remove("Orange");  
    fruits.removeAt(0);  
    print(fruits);  
    String fruit;  
    for (fruit in fruits) {  
        print(fruit);  
    }  
}
```

```
[Apple, Strawberry, Orange, Banana, Grapes]  
[Strawberry, Banana, Grapes]  
Strawberry  
Banana  
Grapes
```

4. Mapping

```
void main() {  
    Map marks = {  
        'Math': 80,  
        'English': 50,  
        'Islamiyat': 70,  
    };  
    print("Original list: $marks");  
    marks["Computer"] = 56;  
    print("Marks after adding subject: $marks");  
}
```

```
marks["English"] = 65;
print("Marks after updating English: $marks");
marks.remove("Islamiyat");
print("Marks after removing: $marks");
print(marks.length);
print(marks.values);
marks.forEach((subject, grade) {
    print("$subject: $grade");
});
}
```

```
Original list: {Math: 80, English: 50, Islamiyat: 70}
Marks after adding subject: {Math: 80, English: 50, Islamiyat: 70,
    Computer: 56}
Marks after updating English: {Math: 80, English: 65, Islamiyat: 70,
    Computer: 56}
Marks after removing: {Math: 80, English: 65, Computer: 56}
3
(80, 65, 56)
Math: 80
English: 65
Computer: 56
```

5. Arithmetic Operators

```
void main() {
    int a = 5;
    int b = 4;
    print(a + b);
    print(a - b);
    print(a * b);
    print(a / b);
    print(a % b);
}
```

```
9
1
20
1.25
1
```

6. Comparison Operators

```
void main() {
    int a = 5;
    int b = 4;
    print(a == b);
    print(a != b);
    print(a > b);
    print(a < b);
    print(a >= b);
    print(a <= b);
}
```

```
false
true
true
false
true
false
```

7. Conditional Statements — If-Else

```
void main() {
    int marks = 75;
    if (marks > 90) {
        print("Grade A+");
    } else if (marks > 80) {
        print("Grade A");
    } else if (marks > 70) {
        print("Grade B");
    } else if (marks > 60) {
        print("Grade C");
    } else if (marks > 50) {
        print("Grade D");
    } else {
        print("You are fail");
    }
}
```

Grade B

8. Switch Statement

```
void main() {
    String day = "Wednesday";
    switch (day) {
        case "Monday":
        case "Tuesday":
        case "Wednesday":
        case "Thursday":
        case "Friday":
            print("Weekday");
            break;
        case "Saturday":
        case "Sunday":
            print("Holidays");
            break;
        default:
            print("Invalid day");
    }
}
```

Weekday

9. For Loop

```
void main() {
    for (int i = 1; i <= 5; i++) {
```

```
print(i);  
}  
}
```

```
1  
2  
3  
4  
5
```